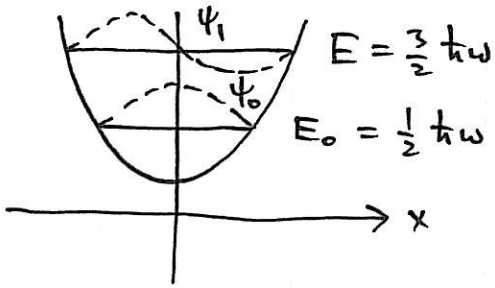
P1 Two $s = 1/2$ particles in harmonic potential



← Single-particle states

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}$$

$$\psi_1(x) = \left[\frac{4}{\pi}\left(\frac{m\omega}{\hbar}\right)^3\right]^{1/4} x\, e^{-m\omega x^2/2\hbar}$$

(a) Total wavefunction must be antisymmetric: lowest energy achieved when both particles are in $\psi_0(x)$ state but their spins form <u>singlet</u>

$$\psi(x_1, x_2) = \psi_0(x_1)\,\psi_0(x_2)\,\underbrace{\frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right)}_{\text{spin singlet}}.$$

$E = 2\cdot\frac{1}{2}\hbar\omega = \hbar\omega$

(b) Total $S=1$ corresponds to triplet: $|1, m\rangle$

$m = +1, 0, -1$

$$|1,+1\rangle = |\uparrow\uparrow\rangle \;;\; |1,0\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\right)$$

$$|1,-1\rangle = |\downarrow\downarrow\rangle$$

$|1,m\rangle$ is symmetric w.r.t. particle exchange

$\Rightarrow$ coordinate wave function must be <u>antisymmetric</u>

$$\Rightarrow \psi(x_1, x_2) = \frac{1}{\sqrt{2}}\left[\psi_0(x_1)\,\psi_1(x_2) - \psi_0(x_2)\,\psi_1(x_2)\right] |1,m\rangle$$

(3 wavefunctions in total)

$$E = \frac{1}{2}\hbar\omega + \frac{3}{2}\hbar\omega = 2\hbar\omega.$$

P2 Calculate $\langle (x_1 - x_2)^2 \rangle$.

To connect with §5.1.2 of Griffiths, denote $\psi_0(x) \to \psi_a(x)$

$\psi_1(x) \to \psi_b(x)$

(a) distinguishable particles:

$$\langle (x_1 - x_2)^2 \rangle_d = \langle x^2 \rangle_a + \langle x^2 \rangle_b - 2 \langle x \rangle_a \langle x \rangle_b$$

where $\langle O \rangle_{\begin{bmatrix} a \\ b \end{bmatrix}} = \int dx \; O(x) \left| \psi_{\begin{bmatrix} a \\ b \end{bmatrix}}(x) \right|^2$

Harmonic oscillator: $\psi_a(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}$

$$\psi_b(x) = \left[\frac{4}{\pi}\left(\frac{m\omega}{\hbar}\right)^3\right]^{1/4} x \, e^{-m\omega x^2/2\hbar}$$

$$\langle x^2 \rangle_a = \int_{-\infty}^{\infty} dx \; x^2 \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{-\frac{m\omega x^2}{\hbar}} = \left(x = \sqrt{\frac{\hbar}{m\omega}}\, y\right) =$$

$$= \sqrt{\frac{\hbar}{m\omega}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \frac{\hbar}{m\omega} \underbrace{\int_{-\infty}^{\infty} dy \; y^2 e^{-y^2}}_{\frac{\sqrt{\pi}}{2}} = \frac{\hbar}{2m\omega}$$

$$\langle x^2 \rangle_b = \left[\frac{4}{\pi}\left(\frac{m\omega}{\hbar}\right)^3\right]^{1/2} \int_{-\infty}^{\infty} dx \; x^2 \cdot \left(x e^{-\frac{m\omega x^2}{2\hbar}}\right)^2 =$$

$$= \left[4\pi^2 \left(\frac{m\omega}{\pi\hbar}\right)^3\right]^{1/2} \int_{-\infty}^{\infty} dx \; x^4 e^{-\frac{m\omega x^2}{\hbar}} =$$

$$= \left[4\pi^2 \left(\frac{m\omega}{\pi\hbar}\right)^3\right]^{1/2} \cdot \left(\frac{\hbar}{m\omega}\right)^{\frac{1}{2} \cdot 5} \underbrace{\int_{-\infty}^{\infty} dy \; y^4 e^{-y^2}}_{\frac{d^2}{d\alpha^2} \sqrt{\frac{\pi}{\alpha}} \Big|_{\alpha=1}} = \frac{3\sqrt{\pi}}{4}$$

$$\Rightarrow \langle x^2 \rangle_b = \left(\frac{4}{\pi}\right)^{1/2} \left(\frac{m\omega}{\hbar}\right)^{3/2} \left(\frac{\hbar}{m\omega}\right)^{5/2} \frac{3\sqrt{\pi}}{4} =$$

$$= \frac{3\hbar}{2m\omega}$$

Also, $\langle x \rangle_a = \langle x \rangle_b = 0$ because the resulting integrals involve <u>odd</u> functions of x.

$$\Rightarrow \langle (x_1 - x_2)^2 \rangle_d = \frac{\hbar}{2m\omega} + \frac{3\hbar}{2m\omega} - 2 \cdot 0 = \frac{2\hbar}{m\omega}$$

(b) Identical bosons (eq. 5.21)

$$\langle (x_1 - x_2)^2 \rangle_{boson} = \langle (x_1 - x_2)^2 \rangle_d - 2 \left| \langle x \rangle_{ab} \right|^2$$

$$\langle x \rangle_{ab} = \int x \, \psi_a^*(x) \, \psi_b(x) \, dx \quad \text{(eq. 5.20)}$$

$$\langle x \rangle_{ab} = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \left(\frac{4}{\pi} \left(\frac{m\omega}{\hbar}\right)^3\right)^{1/4} \int_{-\infty}^{\infty} x \cdot x \cdot e^{-\frac{m\omega}{\hbar} x^2} dx$$

$$= \left(\frac{4}{\pi}\right)^{1/4} \cdot \frac{1}{\pi^{1/4}} \left(\frac{m\omega}{\hbar}\right) \cdot \left(\frac{\hbar}{m\omega}\right)^{\frac{1}{2} \cdot 3} \underbrace{\int_{-\infty}^{\infty} dy \, y^2 e^{-y^2}}_{\frac{\sqrt{\pi}}{2}}$$

$$= \sqrt{\frac{2}{\pi}} \, \frac{\sqrt{\pi}}{2} \left(\frac{\hbar}{m\omega}\right)^{1/2} = \left(\frac{\hbar}{2m\omega}\right)^{1/2}$$

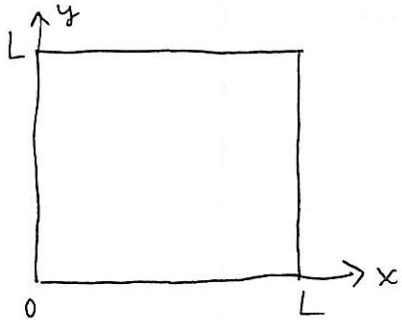
$$\Rightarrow \langle (x_1 - x_2)^2 \rangle_{boson} = \frac{2\hbar}{m\omega} - 2 \frac{\hbar}{2m\omega} = \frac{\hbar}{m\omega}$$

(c) Identical fermions

$$\langle (x_1 - x_2)^2 \rangle_{fermi} = \frac{2\hbar}{m\omega} + 2 \frac{\hbar}{2m\omega} = \frac{3\hbar}{m\omega}$$

Thus $\langle (\Delta x) \rangle_{bose} < \langle (\Delta x) \rangle_d < \langle (\Delta x) \rangle_{fermi}$

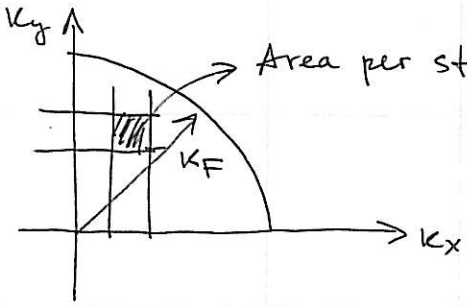
P3 Fermi energy for two-dimensional square well.



Standing waves along X and Y directions:

$$\begin{cases} k_x L = \pi n_x \\ k_y L = \pi n_y \end{cases}$$

$$\Rightarrow E = \frac{\hbar^2 \vec{k}^2}{2m} = \frac{\hbar^2 \pi^2}{2m}\left[\left(\frac{n_x}{L}\right)^2 + \left(\frac{n_y}{L}\right)^2\right], \quad \vec{k} = \frac{\pi}{L}(n_x, n_y)$$



Area per state $= \Delta k_x \cdot \Delta k_y = \frac{\pi}{L} \cdot \frac{\pi}{L} = \frac{\pi^2}{L^2} = \frac{\pi^2}{A}$

where $A = L^2$ is the area of the well.

Each state can be occupied by 2 electrons: with opposite spins $\Rightarrow$

$$\text{number of states} = \frac{1}{4}\left(\pi k_F^2\right)$$

$$\|$$

$$\text{number of electrons}/2 = \frac{N_q}{2}\left(\frac{\pi^2}{A}\right)$$

$$\Rightarrow \frac{\pi k_F^2}{4} = \frac{N_q}{2}\frac{\pi^2}{A} \Rightarrow k_F^2 = 2\pi \frac{N_q}{A}$$

$$k_F = \left(2\pi \frac{N_q}{A}\right)^{1/2}$$

Number of electrons per unit area $\sigma = \frac{N_q}{A} \Rightarrow k_F = (2\pi\sigma)^{1/2}$

$$E_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m}(2\pi\sigma) = \frac{\pi \hbar^2 \sigma}{m}$$

P4 N distinguishable particles in 1d harmonic potential.

$$N = \sum_n e^{-\alpha - \beta E_n}$$

$$E = \sum_n E_n \, e^{-\alpha - \beta E_n}$$

(a) $$N = \sum_n e^{-\alpha - \beta \hbar\omega (n + \frac{1}{2})} = e^{-\alpha - \beta\hbar\omega/2} \sum_{n=0}^{\infty} e^{-\beta\hbar\omega \cdot n} =$$

$$= e^{-\alpha - \beta\hbar\omega/2} \underbrace{\sum_{n=0}^{\infty} \left(e^{-\beta \cdot \hbar\omega}\right)^n}_{\frac{1}{1 - e^{-\beta\hbar\omega}}} = e^{-\alpha} \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}$$

$$\Rightarrow N = e^{-\alpha} \frac{1}{e^{\beta\hbar\omega/2} - e^{-\beta\hbar\omega/2}} = \frac{e^{-\alpha}}{2\sinh(\beta\hbar\omega/2)} \qquad (1)$$

$$E = \sum_{n=0}^{\infty} \hbar\omega\left(n + \frac{1}{2}\right) e^{-\alpha - \beta E_n} = e^{-\alpha} \sum_{n=0}^{\infty} -\frac{d}{d\beta} e^{-\beta E_n} =$$

$$= e^{-\alpha} \left(-\frac{d}{d\beta}\right) \underbrace{\left[\sum_{n=0}^{\infty} e^{-\beta E_n}\right]}_{\frac{1}{2\sinh\left(\frac{\beta\hbar\omega}{2}\right)}}$$

$$\Rightarrow E = e^{-\alpha} \frac{\cosh\left(\beta\frac{\hbar\omega}{2}\right)}{2\sinh^2\left(\beta\frac{\hbar\omega}{2}\right)} \cdot \frac{\hbar\omega}{2} \qquad (2)$$

Using (1) : $e^{-\alpha} = 2\sinh\left(\beta\frac{\hbar\omega}{2}\right) \cdot N$

$$\Rightarrow E = N \cdot \frac{\hbar\omega}{2} \frac{\cosh\left(\beta\frac{\hbar\omega}{2}\right)}{\sinh\left(\beta\frac{\hbar\omega}{2}\right)} = \frac{\hbar\omega}{2} N \frac{e^{\frac{\hbar\omega}{2kT}} + e^{-\frac{\hbar\omega}{2kT}}}{e^{\frac{\hbar\omega}{2kT}} - e^{-\frac{\hbar\omega}{2kT}}}$$

Finally, $E = \frac{\hbar\omega}{2} N \frac{1+e^{-\frac{\hbar\omega}{kT}}}{1-e^{-\frac{\hbar\omega}{kT}}}$ (*)

(b) Quantum limit $\frac{\hbar\omega}{kT} \gg 1 \Rightarrow e^{-\frac{\hbar\omega}{kT}} \to 0$

$E = \frac{\hbar\omega}{2} \cdot N \Rightarrow$ particles sit in the lowest energy state of the oscillator ($E_0 = \frac{1}{2}\hbar\omega$).

(c) Classical limit $\frac{\hbar\omega}{kT} \ll 1 \Rightarrow e^{-\frac{\hbar\omega}{kT}} \simeq 1 - \frac{\hbar\omega}{kT}$

$$\Rightarrow E = \frac{\hbar\omega}{2} N \frac{1+1-\frac{\hbar\omega}{kT}}{1-1+\frac{\hbar\omega}{kT}} = N \frac{\hbar\omega}{2} \frac{2kT}{\hbar\omega} = N \cdot k_B T$$

$\Rightarrow \frac{E}{N} = 2 \cdot \left(\frac{1}{2} k_B T\right)$ [2 degrees of freedom — kinetic + potential — each contributes $\frac{1}{2} k_B T$].

(d) <u>Bonus</u> (partition function)

$$Z = \sum_{n=0}^{\infty} e^{-\beta E_n} = e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} e^{-\beta\hbar\omega n} =$$

$$= \frac{e^{-\beta\hbar\omega/2}}{1-e^{-\beta\hbar\omega}} = \frac{1}{2\sinh\left(\frac{\hbar\omega}{2k_B T}\right)}$$

(I am using $\beta \equiv \frac{1}{k_B T}$ for brevity).

$$F = -k_B T \ln Z = k_B T \ln\left[2\sinh\left(\frac{\hbar\omega}{2k_B T}\right)\right]$$

$$S = \frac{\partial F}{\partial T} = k_B \ln\left[2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right] - \frac{\hbar\omega}{2T} \frac{\cosh\left(\frac{\hbar\omega}{2kT}\right)}{\sinh\left(\frac{\hbar\omega}{2kT}\right)}$$

$\Rightarrow U = F - TS = \frac{1}{2}\hbar\omega \operatorname{cotan}\left(\frac{\hbar\omega}{2k_B T}\right)$ per particle $\Rightarrow$ the same as (*).